

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S1
DATE PERFORMED:	07-16-26	**DATE SUBMITTED:**	
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COMPUTER ENGINEERING

QUEZON CITY

ACTIVITY NO. 2 [ARRAYS, POINTERS AND DYNAMIC MEMORY ALLOCATION]

1. PROCEDURE OUTPUT

Copy and paste the questions from the lab and include your answers after.
After this module, the student should be able to:
a. Implement static and dynamic memory allocation
b. Create dynamically allocated objects using pointers and arrays
c. Solve programming problems using dynamic memory allocation, arrays and pointers

OUTPUT

HOA_1 > dOLOSO > main6.cpp > ...

```cpp
#include <iostream>
#include <string>

class Student{
private:
std::string studentName;
int studentAge;

public:
//cons
Student (std::string newName = "John Doe" , int newAge = 18){
    studentName = std:: move(newName);
    studentAge = newAge;
    std::cout << "Constructor called." << std::endl;
};
//decons
~Student()  {
    std::cout << "Destructor Called" << std::endl;
};
//copy cons
Student(const Student &copyStudent){
    std::cout<< "Copy Contructor Called" << std::endl;
    studentName = copyStudent.studentName;
    studentAge = copyStudent.studentAge;
}
//Display Attributes
void printDetails(){
    std::cout<< this->studentName << "   " << std::endl;
}
```

```
31  };
32
33  int main() {
34  Student student1("DREW", 20);
35  Student student2(student1);
36  Student student3;
37  student3 = student2;
38  return 0;
39  }
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\TIPQC\Documents\CPE010_Doloso_CPE21S1>  & 'c:\Users\TIPQC\.vscode\ex
--stdout=Microsoft-MIEngine-Out-zsiwyojb.evz' '--stderr=Microsoft-MIEngine-Error
Constructor called.
Copy Contructor Called
Constructor called.
Destructor Called
Destructor Called
Destructor Called
PS C:\Users\TIPQC\Documents\CPE010_Doloso_CPE21S1>
```

Initial Driver Program

```
32
33  int main() {
34  const size_t j = 5;
35  Student studentList[j] = {};
36  std::string namesList[j] = {"dane", "gela", "ralph", "drew", "jarel"};
37  int ageList[j] = {15, 16, 18, 19, 16};
38  return 0;
39  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Constructor called.
Constructor called.
Constructor called.
Destructor Called
Destructor Called
Destructor Called
Destructor Called
Destructor Called
PS C:\Users\TIPQC\Documents\CPE010_Doloso_CPE21S1>
```

Modified Driver Program with Student Lists

```
33
34  int main() {
35  const size_t j = 5;
36  Student studentList[j] = {};
37  std::string namesList[j] = {"dane", "gela", "drew", "ralph", "jarel"};
38  int ageList[j] = {15, 16, 18, 19, 16};
39
40  for(int i = 0; i < j; i++){ //loop A
41  Student *ptr = new Student(namesList[i], ageList[i]);
42  studentList[i] = *ptr;
43  }
44
45  for(int i = 0; i < j; i++){ //loop B
46  studentList[i].printDetails();
47  }
48
49  return 0;
50  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Constructor called.
Constructor called.
dane
gela
drew
ralph
jarel
Destructor Called
Destructor Called
Destructor Called
Destructor Called
Destructor Called
PS C:\Users\TIPQC\Documents\CPE010_Doloso_CPE21S1>
```

Observation

On this HOA_2.1

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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